VIETNAM NATIONAL UNIVERSITY, HANOI

UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Hoang Duc Bach

DESIGN AND IMPLEMENTATION OF AN INTENT-BASED META DEX AGGREGATOR WITH COINCIDENCE-OF-WANTS MATCHING ON SUI BLOCKCHAIN

BACHELOR THESIS

Major: Information Technology

HA NOI - 2026

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Supervisor: Dr. Nguyen Thi Hau

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Acknowledgments

I would like to express my sincere gratitude to my supervisor, Dr. Nguyen Thi Hau, for her guidance, valuable feedback, and continuous support throughout this research. Her insights and suggestions greatly helped improve the quality of this thesis.

I also gratefully acknowledge the support of the Sui and Deepbook teams for their generous provision of testnet token funding, which made the practical implementation of this research possible. I extend equal thanks to the broader Sui community for their open and collaborative spirit — their technical discussions, shared resources, and willingness to engage with emerging research have enriched this work in ways that extend well beyond what any single reference could capture.

I am equally thankful to my family, whose steady encouragement and unwavering belief in me sustained me through the most demanding stages of this work.

This thesis is, in many ways, a reflection of the collective generosity of those who supported me — academically and personally. To each of them, I am sincerely grateful.

Declaration

I, Hoang Duc Bach, declare that this thesis is my own original work, composed independently under the supervision of Dr. Nguyen Thi Hau. It has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution.

All sources consulted in the preparation of this thesis have been fully acknowledged and cited in accordance with academic convention. I accept full responsibility for the integrity of this work and for any consequence arising from a failure to meet this declaration, in accordance with the regulations of the University of Engineering and Technology, Vietnam National University.

Undergraduate student,

Hoang Duc Bach

Abstract

As the decentralized finance (DeFi) landscape continues to evolve, slippage and Maximal Extractable Value (MEV) remain significant hurdles for users on Ethereum and EVM-compatible chains. While CoW DEX solutions have successfully mitigated these issues in the EVM space, the Sui network presents a unique environment where its Object-Centric model inherently provides robust protection against certain forms of MEV. Nevertheless, slippage continues to be a persistent challenge for decentralized exchanges (DEXs) on the Sui network.

This thesis focuses on the analysis, architectural design, and development of a CoW DEX on Sui, leveraging the network's object-oriented paradigm and inspired by the core architecture of the Ethereum CoW Protocol. The research proposes a novel auction mechanism incorporating a Commit-Reveal phase and utilizes the Hot Potato Pattern to ensure atomicity, fairness, and the prevention of solver fraud. The system aims to establish an intent-based DEX aggregator that delivers optimal pricing and zero slippage for coincident orders (Coincidence of Wants), while effectively harnessing Sui's core liquidity layers, specifically Deepbook. This work contributes a scalable and secure trading framework that enhances transparency and capital efficiency within the Sui ecosystem.

Keywords: *Blockchain, Decentralized Finance, Coincidence of Wants, Slippage, Maximal Extractable Value, Intent-based, Object-Centric, Sui, Ethereum, Deepbook, CoW Protocol.*

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Chapter 1

Introduction

1.1 Motivation

The emergence of blockchain technology has catalyzed a fundamental transformation in financial infrastructure, giving rise to a new class of decentralized financial systems that operate without traditional intermediaries. As of early 2026, global DeFi Total Value Locked (TVL) stands at approximately $125 billion — a figure that, despite a seasonal correction from the mid-2025 peak of $2.6 billion on Sui alone, reflects the enduring adoption and long-term growth trajectory of on-chain finance. Within this ecosystem, decentralized exchanges (DEXs) have emerged as the primary venue for permissionless asset trading, sustaining an average global daily volume of approximately $19 billion throughout 2025 and into 2026, with Sui-native DEXs contributing approximately $250 million daily [1].

As highlighted in the OKX State of DEXs 2025 Report, the central promise of decentralized trading is the delivery of low slippage and best-price execution. Since the inception of DEXs in 2018, successive architectural generations have pursued this goal through distinct mechanisms. Automated Market Makers (AMMs) democratized liquidity provision but introduced price impact proportional to trade size. Order book DEXs improved price precision but required active market participation and remained susceptible to liquidity fragmentation. DEX Aggregators improved execution by routing across multiple venues, yet remained fundamentally constrained by the depth of underlying pools. Hybrid models combine these approaches at the cost of architectural complexity. Each generation represents an incremental improvement; none fully

resolves the slippage problem under fragmented liquidity conditions.

It is in this context that Maximal Extractable Value (MEV) emerges as a critical phenomenon. MEV refers to the profit that block producers and sophisticated market participants can extract by strategically reordering, including, or excluding transactions within a block prior to finalization [2]. This is made possible by the public visibility of pending transactions in the mempool: before a trade is confirmed, its parameters are observable by any network participant. Searchers — automated agents monitoring the mempool — exploit this transparency to execute three primary forms of MEV extraction. In **front-running**, a searcher identifies a profitable pending trade and submits an identical transaction with a higher priority fee, ensuring execution ahead of the original user and capturing the resulting price movement. In a **sandwich attack**, the attacker places a buy order immediately before the target transaction and a sell order immediately after, artificially inflating the execution price for the victim while profiting from the induced price delta. In **back-running**, the attacker inserts a transaction directly following a price-moving event to capture the resulting arbitrage opportunity before the market equilibrates [2].

The implications of MEV extend beyond individual transaction losses. At scale, MEV introduces systemic inefficiencies: it increases effective transaction costs for ordinary users, distorts price formation on DEXs, and creates an asymmetric information advantage for sophisticated actors with access to specialized infrastructure. These effects collectively undermine the fairness and accessibility that decentralized finance purports to guarantee [2].

In response to MEV's growing impact, multiple mitigation approaches have emerged. Flashbots introduced private transaction relaying to reduce mempool exposure; ZeroMEV developed analytics tooling to quantify and attribute extracted value; and CoW Protocol introduced a structurally distinct execution model. Proposed in 2021, CoW Protocol's **Coincidence of Wants** mechanism batches user trade intents and searches for peer-to-peer matches — settling directly between counterparties before routing residual volume to external liquidity sources. When a user intending to swap token A for token B is matched with a counterparty intending the inverse trade, both orders are settled at the reference market price with zero AMM fee and zero price impact. This batch auction model simultaneously eliminates MEV exposure, reduces slippage, and recovers surplus for users rather than extractors. According to on-chain

data from Dune Analytics, even on Ethereum the average CoW batch contains approximately 1.5 intents, illustrating that peer-to-peer matching remains probabilistic and volume-dependent [3].

On Sui, MEV is addressed through a different set of architectural mechanisms — including priority gas auctions (PGAs) and consensus amplification via SIP-45 — with the explicit design principle that user transaction protection takes priority over extracted value [4]. As a result, the MEV problem that primarily motivates CoW on Ethereum is partially mitigated by Sui's protocol design. Nevertheless, the CoW model retains a powerful and distinct advantage: the ability to achieve **zero slippage** when sufficient matching volume exists. Furthermore, a CoW-style protocol on Sui does not compete with or displace existing DEXs. Rather, it functions as a **meta-DEX aggregator** that discovers and exploits better pricing across Cetus, Aftermath, DeepBook, and other Sui protocols — reducing liquidity fragmentation rather than adding to it. As confirmed by the Sui Q2 2025 DeFi Roundup [1], AMMs and aggregators currently dominate Sui DeFi volume, and no major CoW-style batch auction protocol exists on the network, representing a clear gap in the ecosystem.

Understanding the behavior, scale, and distribution of MEV — as well as the effectiveness of countermeasures such as CoW-style batch auctions — has become essential for designing fair and efficient decentralized trading systems. However, existing implementations of CoW Protocol on Ethereum rely on off-chain infrastructure for solver coordination, intent relay, and price verification, reintroducing trusted intermediaries whose failure modes have been demonstrated in practice: a solver network exploit resulted in direct fund losses, and a separate incident in which the off-chain permission model caused a user to lose nearly $50 million in a misrouted trade — both illustrating the systemic risk of off-chain trust assumptions [5].

This thesis addresses the gap between the theoretical promise of the CoW model and its current implementation constraints. Leveraging Sui's object-centric architecture — which provides native primitives for shared object ownership, programmable transaction blocks, and sub-second finality — this work designs and implements a trustless CoW batch auction protocol in which the entire lifecycle, from intent locking through solver competition to atomic on-chain settlement, is enforced by smart contract logic with no off-chain coordinator. The result is a meta-DEX aggregator that complements Sui's existing liquidity ecosystem, reduces fragmentation, and provides

users with best-execution guarantees backed by on-chain price verification.

1.2 Problem Statement

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1.3 Research Objectives

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1.4 Contributions

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the language.

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Firstname lastname

Chapter 2

Related Works

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2.1 This is section one

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2.2 This is section two

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Chapter 3

System Analysis and Design

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3.2 This is section two

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Appendix A

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Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

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$$\int_0^\infty e^{-\alpha x^2}\mathrm{d}x = \frac{1}{2}\sqrt{\int_{-\infty}^{\infty} e^{-\alpha x^2}\mathrm{d}x \int_{-\infty}^{\infty} e^{-\alpha y^2}\mathrm{d}y} = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}$$

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language. There is no need for special content, but the length of words should match the language.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n\to\infty} \sum_{k=0}^{n} a_0 q^k = \lim_{n\to\infty} a_0 \frac{1-q^{n+1}}{1-q} = \frac{a_0}{1-q}$$

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$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

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$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

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Appendix B

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$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$$

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$$\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$$

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$$a\sqrt[n]{b} = \sqrt[n]{a^n b}$$

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